



Installation and Troubleshooting

Preparation before installation

1. Download docker tar file from:

http://3.210.235.121/l7ammune_docker.v2.360.tar

Username: **l7login**

Password: **L091zKu**

Command:

- `wget --user=l7login --password=L091zKu http://3.210.235.121/l7ammune_docker.v2.360.tar`

2. Check that **/home/l7admin** and **/home/dbpost** directories exist and create them if not.

Commands:

- `mkdir /home/l7admin`
- `mkdir /home/dbpost`

3. Set 755 permissions to **/home/dbpost** and **/home/l7admin**

Commands:

- `chmod -R 755 /home/dbpost`
- `chmod -R 755 /home/l7admin`

4. Check if **docker** package installed, if not install:

<https://docs.docker.com/engine/install/ubuntu/>

5. Download helper scripts **start**, **stop**, **enter** from: <http://3.210.235.121/start>, <http://3.210.235.121/stop>, <http://3.210.235.121/enter>

Commands:

- `wget --user=l7login --password=L091zKu http://3.210.235.121/start`
- `wget --user=l7login --password=L091zKu http://3.210.235.121/stop`
- `wget --user=l7login --password=L091zKu http://3.210.235.121/enter`

Explanation:

- **start** stops the currently running container (if it exists), cleans all its volumes and runs a new container.
- **stop** stops the currently running container (if it exists) and cleans all its volumes.
- **enter** opens a shell from inside the container.

6. Copy scripts to **/sbin** directory and set execution permissions.

Commands:

- `cp <script_name> /sbin/<script_name>`
- `chmod +x /sbin/<script_name>`



Installation and Troubleshooting

Installation

1. Load .tar file as a docker image.

Command:

- docker load <I7ammune_docker.v2.360.tar

2. Start the docker container.

Command:

- start

3. Wait a few minutes and access Ammune UI - https://<Ammune_IP>:8443/

4. Proceed to enter license key:

DDcarStAZ5QE7T4r1L/C9IER4Wa4vVrccYABP5XTb03UMTAehO6nP3ziAeXr8Y4M6sh1w/qMVcg
GKCKtRViGkbPNdP4qNC5Fh6plgQgbqLpTxmOPOz6tducSr30auo4sc8NSMhniQUsOvBUNsf/3
M1v5HECdqakzyAfn6AhpjJturV+cxLhDxx4lgEh0M3sAx3xTMgX221UKrI5MH6KURJYnmwvjN
CdD4r8kqZlqRelC4/RLMzZMa5Z+s0WlfNB00RixxCGXROr5a0ixYqolQ==



Installation and Troubleshooting

5. Create an **admin**:

1 CREATE ADMIN 2 SELECT MODE 3 CONFIGURE DOMAINS

Create your first account

Username
maxim

Email
maxim@i7defense.com

Password

Confirm Password

NEXT

6. Select mode **Behind Load Balancer**:

SELECT MODE

Please pick the way your instances are connected - Behind Load Balancer or directly to DNS

Behind Load Balancer

7. Click on **Add Domain**:

Domain	Front IP	Domain Protocol	Domain Port	Protected Instance IP	Protected Instance Protocol	Protected Instance Port	Add
	SELECT	SELECT			SELECT		+

BACK ADD DOMAIN

8. Click on **Finish**:

Domain	Front IP	Domain Protocol	Domain Port	Protected Instance IP	Protected Instance Protocol	Protected Instance Port	Add
	SELECT	SELECT			SELECT		+

BACK FINISH

Now you should be able to see the following screen:



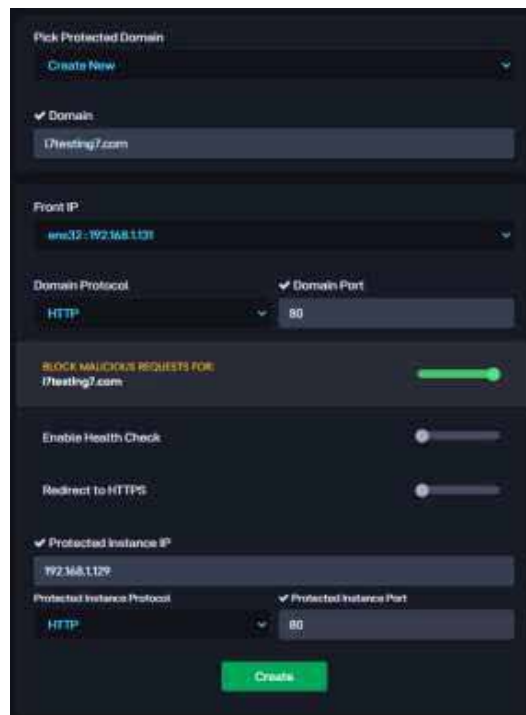
Installation and Troubleshooting



If you do, you have successfully installed the Ammune.ai platform!

Configuration - Inline Mode - All in One Mode

1. Go to **Settings** -> **Proxy Settings**
2. Configure a new protected domain with parameters that the client gives, for example:



3. For sanity check, you can run:
 - curl -H "Host: l7testing7.com" http://<Ammune_IP>/
 - curl -H "Host: l7testing7.com" http://<Ammune_IP>?l7test=../ Wait 30 seconds and check in **UI -> API-WAF -> Real Time Dashboard** that a normal request and alert pop up.



Installation and Troubleshooting

Configuration - Inline Mode - Management Only Mode

1. Go to **Settings** -> **Global Settings** -> **System Mode** -> **Management Only** and commit.
2. Proceed to do steps 1-3 in the previous paragraph Inline Mode - All in One Mode.

Configuration - Inline Mode - Engine Only Mode

1. Go to **Settings** -> **Global Settings** -> **System Mode** -> **Engine Only**, fill the management machine's IP and commit.

Configuration - Monitoring Mode

1. Go to **Settings** -> **Global Settings** -> **Engine Mode** -> **Monitor**
2. Go to **Settings** -> **Monitoring Settings**, fill the protected IP ranges and ports that the client provides and commit.

Troubleshooting

1. If the configuration mode is **Inline & All in One**, and something goes wrong (for example, login screen doesn't show up, or attacks to Ammune are not blocked, etc), check that the next ports are up and occupied by Ammune processes:
 - 10 Analytics ports: **7777, 8888, 18000-18007**
 - 8 Agent ports: **50020-50027**
 - 3 License Manager ports: **16161, 17171, 18181**
 - 1 L7_WAF_RPC port: **26000**
 - 4 Nginx ports: **389, 5300, 8443, 26001**
 - 1 PostgreSQL port: **5432**
 - 3 Python scripts ports: **5557, 19002, 60031**

To check used ports, enter the container with the command:

- enter

Then run command:

- netstat -tulpn

to see all of the ports that are currently occupied.

2. If the configuration mode is **Inline & Management Only**, and something goes wrong, check that the next ports are up and occupied by Ammune.ai processes:
 - 10 Analytics ports: **7777, 8888, 18000-18007**
 - 3 License Manager ports: **16161, 17171, 18181**
 - 1 L7_WAF_RPC port: **26000**
 - 5 Nginx ports: **389, 5300, 8443, 9090, 26001**
 - 1 PostgreSQL port: **5432**
 - 4 Python scripts ports: **5555, 5557, 19002, 60031**



Installation and Troubleshooting

To check used ports, enter the container with the command:

- enter

Then run command:

- netstat -tulpn

to see all of the ports that are currently occupied.

3. If the configuration mode is **Inline & Engine Only**, and something goes wrong, check that the next ports are up and occupied by Ammune processes:

- 8 Agent ports: **50020-50027**
- 3 License Manager ports: **16161, 17171, 18181**
- 1 L7_WAF_RPC port: **26000**
- 16 Nginx ports: **389, 5300, 5555, 7777, 8443, 8888, 18000-18007, 19001, 26001**
- 1 PostgreSQL port: **5432**

To check used ports, enter the container with the command:

- enter

Then run command:

- netstat -tulpn

to see all of the ports that are currently occupied.

Also, check that port **9090** is occupied in the **Management Only Ammune** machine with the ip that you filled in **Settings** -> **Global Settings**

To check if a port is up in another machine, run one of the commands:

- telnet <management_ip> 9090
- nc -zv <management_ip> 9090

4. If the configuration mode is **Monitor**, and something goes wrong, check that the next ports are up and occupied by Ammune processes:

- 10 Analytics ports: **7777, 8888, 18000-18007**
- 8 Agent ports: **50020-50027**
- 3 License Manager ports: **16161, 17171, 18181**
- 1 L7_WAF_RPC port: **26000**
- 4 Nginx ports: **389, 5300, 8443, 26001**
- 1 PostgreSQL port: **5432**
- 3 Python scripts ports: **5557, 19002, 60031**

To check used ports enter the container with command:

- enter

Then run command:

- netstat -tulpn

to see all of the ports that are currently occupied.



Installation and Troubleshooting

5. After running the start command in the installation, if you can't access the UI and the wizard doesn't show up, the **Nginx** might not have started. To check it enter the container with the command:
 - enter

Then run command:

- ps aux | grep nginx

If no results show up, the Nginx is down.

Run:

- nginx -t

To check the Nginx configuration state.

If there is a problem with configurations, try to fix it and then wait for Nginx Watchdog to start it again (preferably) or start Nginx manually with the command:

- nginx

Another issue might be that the port **8443**, which the UI sits on, could be occupied by another process, in this case, kill the other process and wait for the watchdog to run, Nginx, or try to start it manually like mentioned above.

6. If some time after the installation and configuration, you or the client try to login and the login fails (even though the credentials are correct), the **PostgreSQL DB** might be down.

To check it run:

- ps aux | grep Engine_Database
- ps aux | grep wafdb

If there are no results for either command, then **PostgreSQL** is down.

Try to run:

- service postgresql start

And check if the **DB** is down again.

If it's down, then there might not be enough free space for the **DB** in the disk, try to free up some space and run again:

- service postgresql start

If it's still down, maybe there are no correct permissions in **/home/dbpost**, run:

- chmod -R 755 /home/dbpost
- service postgresql start